Intelligent System →

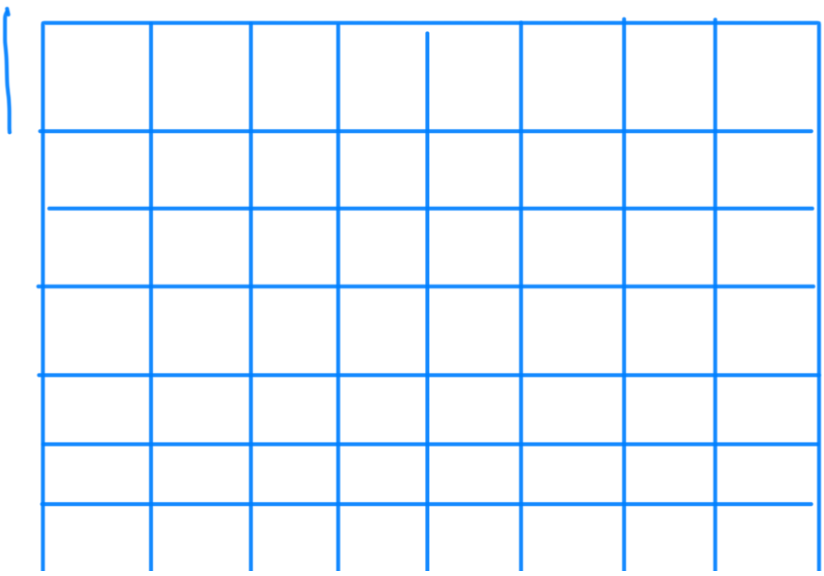
Problem, Problem space, search

(i) Define the problem precisely
(ii) Analyze problem
(iii) Isolate & represent the task knowledge
(iv) Choose the best problem solving Technique

10^{120} rules in chess

Define problem as a state space search

(1) Initial state
(2) Goal state
(3) legal moves



while (Pawn)

Square (File e, rank 2)

AND

Square (File e, rank 3)

AND

Square (File e, rank 4)

is empty

Water Jug problem

2 Jugs 4 gallon

3 gallon

No measuring markers

Goal → 2 gallon water in 4 gallon Jug.

State space

(x, y)

$x = 1, 2, 3, 4$

$y = 1, 2, 3$.

Start state → $(0, 0)$

Goal state → $(2, n)$

Production rule → of Water Jug Problem

① (x, y)

if $x < 4 \rightarrow (4, y)$ Fill the 4 gallon jug.

② if (x, y)

if $x < 3 \rightarrow (x, 3)$

③ (x, y)

$x > 0 \rightarrow (x - d, y)$

Pour some water from 4 gallon jar.

④ (x, y)

If $y > 0 \rightarrow (x, y - d)$

Pour some water from 3 gallon jar.

⑤ (x, y)

$x > 0 \rightarrow (0, y)$ Empty 4

⑥ $(x, y) \rightarrow (x, 0)$ Empty 3

$y > 0$

⑦ (x, y)

if $(x+y \geq 4$ and $y>0)$

$(4, (y-(4-x)))$

8

12